

LIFEWS AgriShine(TM) GrowMeal(TM)

SCHOOL IMPLEMENTATION & GARDEN OPERATIONS MANUAL

For Principals, School Coordinators, Teachers & Garden Managers

Purpose	Institutional operating standard for launching, managing, monitoring and improving the GrowMeal school garden learning program.
Scope	Nursery, Primary 1-3, Primary 4-6, JSS 1-3 and SSS 1-3.
Operating principle	Install only what the school can safely teach with, water, supervise, maintain and document.
Instructional link	Operations support the GrowMeal 12-week teacher plans and the complete student/teacher resource system.

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1. SCHOOL ONBOARDING & READINESS

Leadership approval	Appoint a school GrowMeal lead and confirm participating classes, responsible teachers, garden-maintenance responsibility and reporting line.
Readiness review	Confirm safe space, water access, learner access, storage, timetable capacity, basic tools, teaching materials and maintenance arrangements before launch.
Baseline record	Photograph/map the proposed site, record existing infrastructure and constraints, and identify which garden systems can be supported reliably.
Launch condition	Do not install more garden infrastructure than the school can safely water, supervise and maintain.

2. SELECTING GARDENS BY LEVEL

Nursery	Use small, highly visible and accessible gardens emphasizing letters, numbers, shapes, leaves, pollinators, reused containers, micro-growing and discovery.
Primary 1-3	Add food-production, raised-bed, wall, sack/growbag and ground-production experiences while retaining observation and simple measurement.
Primary 4-6	Introduce experimental grids, keyhole/compost systems, gravity drip, rotation, water-smart and introductory soilless learning.
JSS 1-3	Use production, systems, rotation/intercropping, compost/soil, soilless, sensor-ready and applied systems gardens for structured investigations.
SSS 1-3	Use advanced trials, controlled experiments, irrigation engineering, water productivity, instrumentation, smart irrigation, research/design and integrated food-system laboratories.

3. SITE PLANNING & ZONING

Learning zone	Provide safe learner gathering space, demonstration position and clear sightlines.
Production zone	Separate active growing beds/containers from main circulation where necessary.
Water zone	Keep water points accessible, controlled and free of standing-water hazards.
Tool/material zone	Provide secure storage and a defined issue/return point.
Food-literacy zone	Where Ingredient Lab or Farm-to-Food work occurs, keep soil tools separate from clean food-contact materials.
Accessibility	Use practical path widths, reachable work heights and alternative stations so learners with different access needs can participate.

4. ROLES & RESPONSIBILITIES

Principal / school leader	Approves implementation, timetable, safeguarding arrangements, budget, staffing and escalation procedures.
GrowMeal coordinator	Maintains the school implementation calendar, garden status, inventory, teacher coordination, records and term reports.
Teacher	Plans and teaches the 12-week learning cycle, supervises learners, assesses evidence and verifies passports.
Garden custodian/team	Maintains water, beds, structures and routine care outside lesson periods as assigned.
Learners	Participate in age-appropriate care, practical work, evidence recording and cleanup under supervision.
Community contributors	Support only through school-approved roles and safeguarding procedures.

5. 12-WEEK OPERATING CALENDAR

Weeks 1-10	One level-specific garden focus per week, with preparation, demonstration, practical activity, journal evidence, mission/application and assessment.
Week 11	Maintenance catch-up, evidence review, remediation, assessment moderation and portfolio/passport completion.
Week 12	Garden learning showcase, demonstrations, reflection, certification and term review.
Between lessons	Maintain essential watering, crop care, system checks and safety controls so learning assets remain functional.

6. DAILY/WEEKLY GARDEN OPERATIONS

Daily/regular check	Water need, obvious plant stress, leaks, damaged structures, trip hazards, unsafe tools/materials and unauthorized access.
Weekly check	Bed/container condition, weeds, mulch, irrigation function, water storage, labels, pathways, tool inventory and learning-station readiness.
Record	Document important failures, repairs, crop losses and unusual observations. Operational problems can become learning data.
Escalate	Close or isolate any area that cannot be made safe immediately.

7. WATER MANAGEMENT

Priority	Reliable water access is a prerequisite for garden selection and scale.
Practice	Use watering schedules suited to the crop/system, avoid unnecessary runoff and inspect hoses, emitters, containers and storage for leaks.
Learner role	Age-appropriate learners may measure, compare and record water use under teacher supervision.
Water quality	Use water appropriate to the intended garden/food-contact activity and follow local school/public-health requirements.

8. TOOLS, EQUIPMENT & INVENTORY

Control	Assign tools to a secure storage location and use an issue/return system.
Inspection	Remove damaged, sharp, loose or otherwise unsafe tools from learner use.
Age suitability	Select tool size and complexity for the learner group; higher-risk tools remain under adult control.
Inventory cycle	Count durable tools at least at the beginning and end of each term; track frequently used consumables separately.
Replacement	Budget for wear, lost labels, broken containers, irrigation fittings, gloves/PPE where required and measurement consumables.

9. INPUTS, SOIL & GROWING MEDIA

Traceability	Know the source and intended use of soil, compost, amendments, fertilizers and other inputs.
Storage	Keep inputs labeled, dry/secure where required and inaccessible to unauthorized learners.
Application	Follow manufacturer labels and school procedures. Do not ask learners to handle hazardous products.
Learning use	Compost, soil and growing-media activities should emphasize observation, measurement and management without compromising hygiene.

10. FOOD SAFETY & INGREDIENT ACTIVITIES

Separation	Keep garden tools/soil-contact containers separate from food-contact trays, utensils and clean containers.
Hand hygiene	Require handwashing before and after approved ingredient handling.
Produce	Use only teacher-approved produce with known handling history. Discard visibly spoiled, contaminated or uncertain material.
Tasting	GrowMeal activities do not require tasting. Any tasting must be separately authorized under school food-safety, allergy and safeguarding procedures.
Higher-risk preparation	Adults control knives, heat, electrical appliances and processing machinery unless a formal school protocol authorizes supervised use.

11. SAFEGUARDING & LEARNER SUPERVISION

Boundaries	Define where learners may work and what stop signal is used.
Ratios	Use group sizes and supervision appropriate to age, task, tools and site complexity.
Visitors	External facilitators operate through school authorization and safeguarding procedures.
Incident handling	Stop work, secure the area, provide/seek appropriate assistance and follow the school's incident reporting process.

12. PROCUREMENT & BUDGETING

Procurement principle	Buy for maintainability, safety and instructional use rather than visual complexity.
Priority order	Water access and safety; durable garden infrastructure; basic tools; learning/measurement materials; replacement consumables; optional advanced equipment.
Local sourcing	Use locally available, repairable materials where they meet safety and performance needs.
Budget categories	Infrastructure, water/irrigation, tools, growing inputs, seeds/planting material, learning resources, measurement equipment, storage, maintenance and contingency.

13. MAINTENANCE & REPAIR

Preventive maintenance	Inspect before failure: fasteners, supports, irrigation joints, containers, paths, storage and labels.
Repair record	Log defect, date, action, responsible person and closure date.
Student participation	Learners may help with safe maintenance tasks when those tasks have a clear learning objective and supervision.
Decommissioning	Remove or isolate systems that are no longer structurally safe, hygienic or maintainable.

14. SEASONAL & HOLIDAY CONTINUITY

Term planning	Select crops and activities that fit the school calendar and likely maintenance capacity.
Holiday plan	Assign watering/care responsibility before closure; reduce planted area when reliable care is unavailable.
Restart	Inspect structures, water systems, growing media and stored tools before learners resume practical work.
Learning continuity	Retain records from previous cycles so crop/system outcomes can be compared across terms.

15. MONITORING & PROGRAM KPIS

Participation	Classes/learners actively using GrowMeal resources and gardens.
Instruction	Planned garden weeks completed and evidence collected.
Competence	Assessment dimension results and passport completion.
Operations	Garden systems functional, water reliability, maintenance closures and repair time.
Safety	Incidents, near misses and corrective actions.
Resource stewardship	Tool losses, consumable use, avoidable crop/material losses and waste-reduction actions.

16. REPORTING

Weekly	Teacher/garden notes: lesson delivered, garden status, maintenance need and safety issue.
Term	Coordinator report: participation, learning evidence, assessment summary, passport completion, garden condition, inventory, incidents, repairs and next actions.
Evidence	Use photographs only under school consent/privacy rules. Prefer traceable records over promotional images as proof of implementation.
Escalation	Urgent safety, water or infrastructure failures should not wait for the term report.

17. PARENT & COMMUNITY ENGAGEMENT

Purpose	Use community engagement to strengthen learning, maintenance knowledge and local food-system understanding.
Boundaries	Do not shift supervision or safeguarding responsibility away from the school.
Activities	Approved garden days, demonstrations, local grower talks, learner exhibitions, ingredient displays and resource-reuse activities.
Communication	Explain that GrowMeal is an educational garden and food-literacy program, not a medical or therapeutic program.

18. QUALITY ASSURANCE & SCHOOL CERTIFICATION

Implementation evidence	12-week records, garden-use evidence, teacher records, assessment moderation, passport verification and operational/safety records.
Quality review	Check whether the school is delivering practical learning, not merely maintaining a decorative garden.
Certification decision	Certification should depend on implementation evidence and safe operation, not garden appearance alone.
Improvement plan	Where standards are incomplete, record corrective actions, responsible persons and review dates.

19. END-OF-TERM CLOSEOUT

Learners	Complete assessment, remediation, passport verification and reflection.
Teachers	Moderate samples, identify curriculum/resource issues and archive key records.
Garden	Harvest/manage remaining crops appropriately, repair systems, prepare holiday maintenance and secure tools.
Leadership	Review KPIs, budget, incidents, maintenance, staffing and next-term scale.

20. EMERGENCY/FAILURE RESPONSE

Water failure	Prioritize essential crop/system protection, identify repair/alternative supply and reduce activities that depend on unavailable water.
Infrastructure damage	Isolate the affected area immediately and reopen only after inspection/repair.
Contamination concern	Stop food-contact/harvest use of affected material until the school determines it is safe.
Severe crop loss	Document cause where possible, remove unsafe material and use the event as evidence for future management decisions.
Missing equipment	Suspend tasks that cannot be performed safely or validly without the required equipment.

APPENDIX A - SCHOOL READINESS CHECKLIST

Check / Item	Status / Qty	Action / Notes
Leadership approval		
GrowMeal coordinator appointed		
Participating classes identified		
Safe garden site		
Reliable water		
Secure tool storage		
Basic tools available		
Learning materials available		
Maintenance responsibility assigned		
Safeguarding procedures confirmed		
Food-safety procedure confirmed		
12-week timetable protected		

APPENDIX B - WEEKLY GARDEN OPERATIONS CHECK

Check / Item	Status / Qty	Action / Notes
Water system functioning		
Beds/containers safe		
Paths clear		
Structures secure		
Labels readable		
Tools accounted for		
Growing media condition checked		
Crop stress/pests observed		
Waste removed/managed		
Learning station ready		
Repair needed recorded		
Area safe to open		

APPENDIX C - TOOL & MATERIAL INVENTORY

Check / Item	Status / Qty	Action / Notes
Hand tools		
Watering equipment		
Irrigation fittings		
Containers/growbags		
Labels/stakes		
Measuring tools		
Clean food-contact trays/containers		
PPE where required		
Seeds/planting material		
Growing inputs		
Learning cards/manuals		
Other		

APPENDIX D - MAINTENANCE & REPAIR LOG

[illegible]

APPENDIX E - TERM IMPLEMENTATION REVIEW

Weeks delivered	_____
Gardens used	_____
Learners participating	_____
Assessment/passport status	_____
Garden condition	_____
Water reliability	_____
Safety issues	_____
Inventory losses	_____
Repairs outstanding	_____
Strongest learning evidence	_____
Priority improvements	_____
Next-term decision	_____